R4 Analysis: Economic Development, Energy Intensity, and CO2 Emissions

Group 9

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1 Research Question 4

1.1 Research Question

Do countries become more energy-efficient as they develop, and how does this affect CO2 emissions over time?

1.2 Motivation

This analysis extends the project beyond prediction and clustering by examining a deeper development question. Instead of only asking whether variables are associated with emissions, this section studies whether economic development is linked to lower energy intensity and whether this, in turn, changes CO2 emissions per capita.

The analysis addresses three specific ideas:

1. As GDP increases, do countries tend to use less energy per unit of output?
2. Is higher energy intensity associated with higher CO2 emissions per capita?
3. Does the effect of GDP on CO2 emissions depend on energy intensity?

These questions matter because they help distinguish whether emissions are driven more by economic scale or by the way energy is used.

2 Data and Variables

```r
library(tidyverse)
library(car)
library(broom)
library(scales)
library(plm)
```

```r
library(knitr)

df <- read.csv("cleaned_co2_data_20vars.csv")

str(df)
```

```
## 'data.frame':    2640 obs. of  20 variables:
##  $ country                   : chr  "Albania" "Albania" "Albania" "Albania" ...
##  $ year                      : int  1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 ...
##  $ iso_code                  : chr  "ALB" "ALB" "ALB" "ALB" ...
##  $ population                : int  3269417 3258573 3245680 3229665 3210134 3188598 3166147
##  $ gdp                       : num  1.06e+10 1.16e+10 1.25e+10 1.10e+10 1.20e+10 ...
##  $ primary_energy_consumption: num  13 14.3 13.5 12.9 14.9 ...
##  $ energy_per_capita         : num  3981 4392 4160 4001 4637 ...
##  $ energy_per_gdp            : num  1.23 1.23 1.08 1.17 1.24 ...
##  $ co2                       : num  1.92 2.09 2.02 1.54 1.75 ...
##  $ co2_per_capita            : num  0.588 0.641 0.622 0.478 0.547 ...
##  $ total_ghg                 : num  7.23 7.3 6.82 5.99 6.22 ...
##  $ consumption_co2           : num  2.37 2.52 2.89 2.19 2.43 ...
##  $ methane                   : num  4.13 4.07 3.88 3.62 3.46 ...
##  $ nitrous_oxide             : num  1.31 1.3 1.17 1.1 1.14 ...
##  $ cement_co2                : num  0.0515 0.1032 0.1049 0.0518 0.0436 ...
##  $ flaring_co2               : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ trade_co2                 : num  0.442 0.434 0.877 0.649 0.678 ...
##  $ coal_co2                  : num  0.1685 0.1612 0.1136 0.044 0.0548 ...
##  $ gas_co2                   : num  0.0989 0.0513 0.044 0.0366 0.0329 ...
##  $ oil_co2                   : num  1.6 1.77 1.76 1.41 1.62 ...
```

```r
summary(df)
```

```
##    country               year        iso_code           population
##  Length:2640        Min.   :1990   Length:2640        Min.   :3.679e+05
##  Class :character   1st Qu.:1999   Class :character   1st Qu.:5.353e+06
##  Mode  :character   Median :2007   Mode  :character   Median :1.503e+07
##                     Mean   :2007                      Mean   :6.801e+07
##                     3rd Qu.:2015                      3rd Qu.:5.741e+07
##                     Max.   :2022                      Max.   :1.426e+09
##       gdp            primary_energy_consumption energy_per_capita
##  Min.   :4.761e+09   Min.   :    4.548          Min.   :   330.9
##  1st Qu.:8.655e+10   1st Qu.:  128.226          1st Qu.: 11999.1
##  Median :2.829e+11   Median :  386.344          Median : 29933.7
##  Mean   :9.478e+11   Mean   : 1525.665          Mean   : 37790.2
##  3rd Qu.:8.098e+11   3rd Qu.: 1187.485          3rd Qu.: 48885.7
##  Max.   :2.697e+13   Max.   :44516.492          Max.   :313668.4
##  energy_per_gdp         co2             co2_per_capita       total_ghg
##  Min.   : 0.2152   Min.   :9.668e-01   Min.   : 0.06412   Min.   :    1.583
```

```
##  1st Qu.: 1.0648   1st Qu.:2.905e+01   1st Qu.: 2.40692   1st Qu.:   53.036
##  Median : 1.4060   Median :6.867e+01   Median : 6.07548   Median :  125.169
##  Mean   : 1.6701   Mean   :3.421e+02   Mean   : 7.39173   Mean   :  502.879
##  3rd Qu.: 2.0236   3rd Qu.:2.776e+02   3rd Qu.: 9.41820   3rd Qu.:  411.836
##  Max.   :17.1433   Max.   :1.171e+04   Max.   :76.71462   Max.   :13446.920
##  consumption_co2        methane          nitrous_oxide         cement_co2
##  Min.   :    0.546   Min.   :   0.1466   Min.   :  0.04315   Min.   :  0.0000
##  1st Qu.:   31.900   1st Qu.:   8.1443   1st Qu.:  2.63618   1st Qu.:  0.7004
##  Median :   88.801   Median :  23.2491   Median :  8.49542   Median :  2.1451
##  Mean   :  341.665   Mean   :  82.3393   Mean   : 26.25617   Mean   : 12.3455
##  3rd Qu.:  274.245   3rd Qu.:  75.7045   3rd Qu.: 23.76059   3rd Qu.:  7.6319
##  Max.   :10427.033   Max.   :1513.4746   Max.   :466.87686   Max.   :828.7100
##   flaring_co2          trade_co2             coal_co2           gas_co2
##  Min.   : 0.000000   Min.   :-1604.4298   Min.   :   0.000   Min.   :   0.000
##  1st Qu.: 0.000056   1st Qu.:   -2.2695   1st Qu.:   1.003   1st Qu.:   3.924
##  Median : 0.321382   Median :    4.5329   Median :  12.001   Median :  15.768
##  Mean   : 3.401049   Mean   :   -0.4565   Mean   : 141.563   Mean   :  66.099
##  3rd Qu.: 2.849402   3rd Qu.:   19.1819   3rd Qu.:  57.900   3rd Qu.:  63.303
##  Max.   :84.519539   Max.   :  573.5951   Max.   :8468.947   Max.   :1707.391
##     oil_co2
##  Min.   :   0.3664
##  1st Qu.:  10.9065
##  Median :  32.5795
##  Mean   : 114.1464
##  3rd Qu.:  87.2593
##  Max.   :2584.1299
```

We use the following variables:

- `gdp`: level of economic development
- `energy_per_gdp`: energy intensity, interpreted as energy used per unit of GDP
- `co2_per_capita`: environmental outcome
- `population`: demographic control
- `energy_per_capita`: direct energy use control
- `country` and `year`: panel identifiers

A lower value of `energy_per_gdp` indicates better energy efficiency, because less energy is required to generate one unit of economic output.

```
df_r4 <- df %>%
  select(country, year, gdp, energy_per_gdp, co2_per_capita, population, energy_per_capita) %>%
  filter(
    !is.na(country),
    !is.na(year),
    !is.na(gdp),
    !is.na(energy_per_gdp),
    !is.na(co2_per_capita),
```

```r
    !is.na(population),
    !is.na(energy_per_capita)
  )

glimpse(df_r4)
```

```
## Rows: 2,640
## Columns: 7
## $ country           <chr> "Albania", "Albania", "Albania", "Albania", "Albania~
## $ year              <int> 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002~
## $ gdp               <dbl> 10596114842, 11600732968, 12468867839, 11020137102, ~
## $ energy_per_gdp    <dbl> 1.2282000, 1.2336622, 1.0828067, 1.1725888, 1.237635~
## $ co2_per_capita    <dbl> 0.5884248, 0.6410981, 0.6215979, 0.4777613, 0.546615~
## $ population        <int> 3269417, 3258573, 3245680, 3229665, 3210134, 3188598~
## $ energy_per_capita <dbl> 3980.571, 4391.918, 4159.798, 4001.062, 4636.601, 36~
```

```r
summary(df_r4)
```

```
##    country               year           gdp             energy_per_gdp
##  Length:2640        Min.   :1990   Min.   :4.761e+09   Min.   : 0.2152
##  Class :character   1st Qu.:1999   1st Qu.:8.655e+10   1st Qu.: 1.0648
##  Mode  :character   Median :2007   Median :2.829e+11   Median : 1.4060
##                     Mean   :2007   Mean   :9.478e+11   Mean   : 1.6701
##                     3rd Qu.:2015   3rd Qu.:8.098e+11   3rd Qu.: 2.0236
##                     Max.   :2022   Max.   :2.697e+13   Max.   :17.1433
##  co2_per_capita       population        energy_per_capita
##  Min.   : 0.06412   Min.   :3.679e+05   Min.   :   330.9
##  1st Qu.: 2.40692   1st Qu.:5.353e+06   1st Qu.: 11999.1
##  Median : 6.07548   Median :1.503e+07   Median : 29933.7
##  Mean   : 7.39173   Mean   :6.801e+07   Mean   : 37790.2
##  3rd Qu.: 9.41820   3rd Qu.:5.741e+07   3rd Qu.: 48885.7
##  Max.   :76.71462   Max.   :1.426e+09   Max.   :313668.4
```

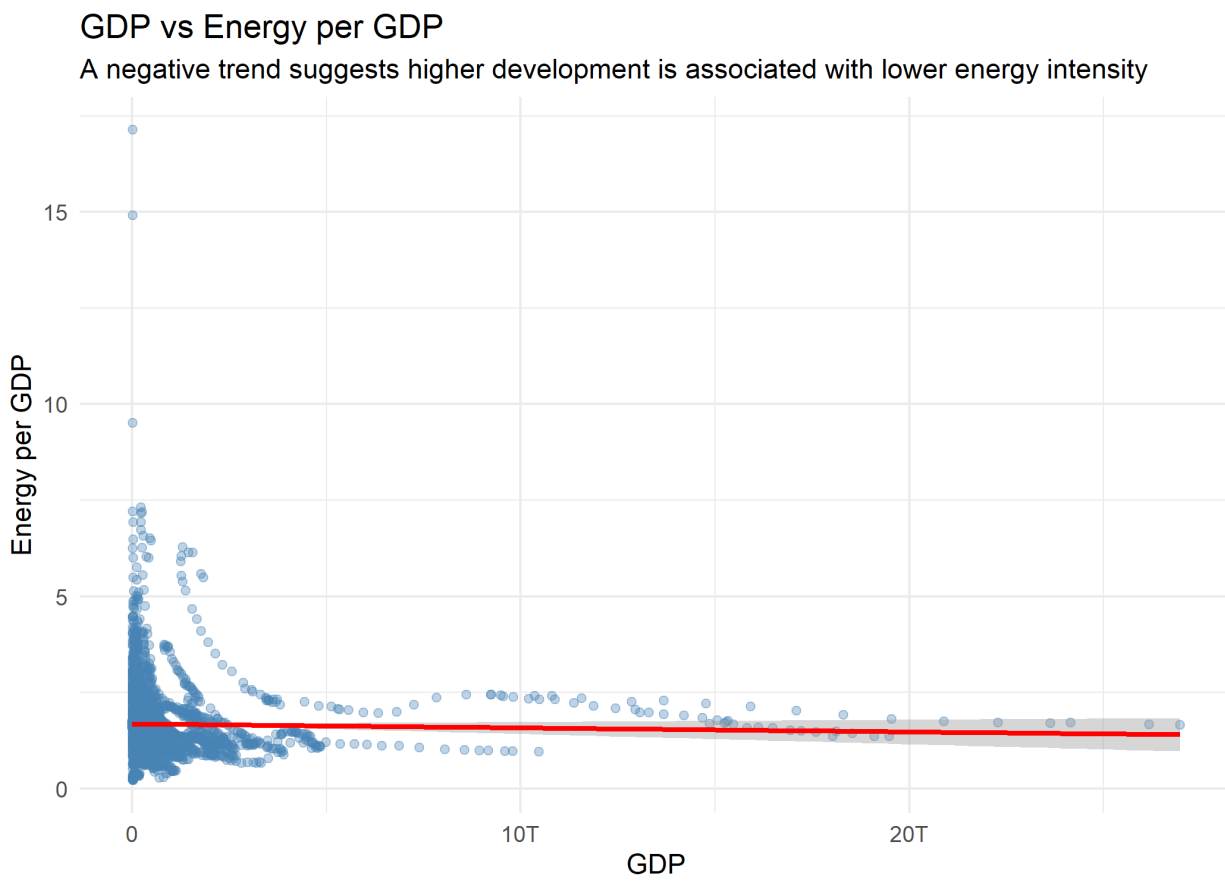
3 Exploratory Data Analysis

3.1 GDP vs Energy per GDP

If countries become more efficient as they develop, energy intensity should decline as GDP increases.

```r
ggplot(df_r4, aes(x = gdp, y = energy_per_gdp)) +
  geom_point(alpha = 0.35, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  scale_x_continuous(labels = label_number(scale_cut = cut_short_scale())) +
  labs(
    title = "GDP vs Energy per GDP",
```

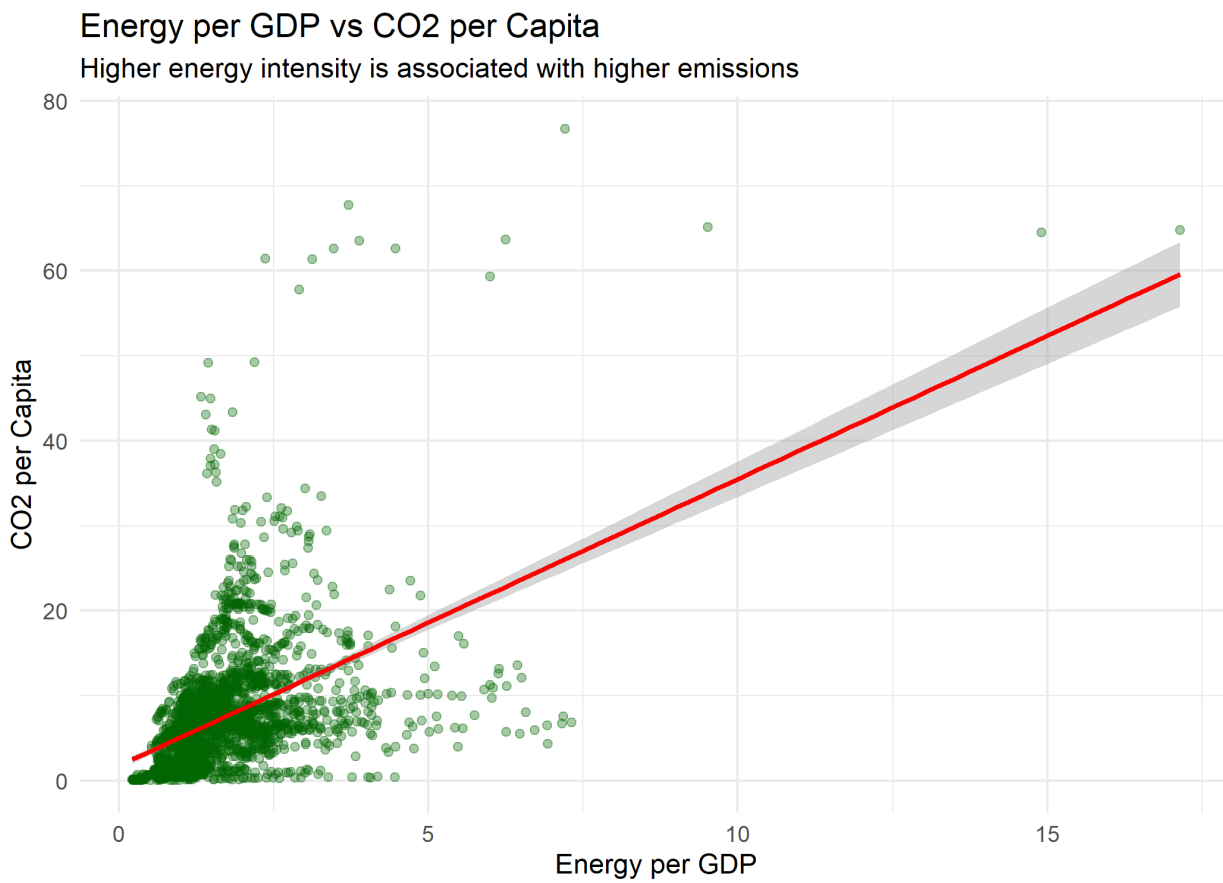
```
    subtitle = "A negative trend suggests higher development is associated with lower energy in
    x = "GDP",
    y = "Energy per GDP"
  ) +
  theme_minimal()
```



3.2 Energy per GDP vs CO2 per Capita

This plot examines whether more energy-intensive economies tend to emit more CO2 per capita.

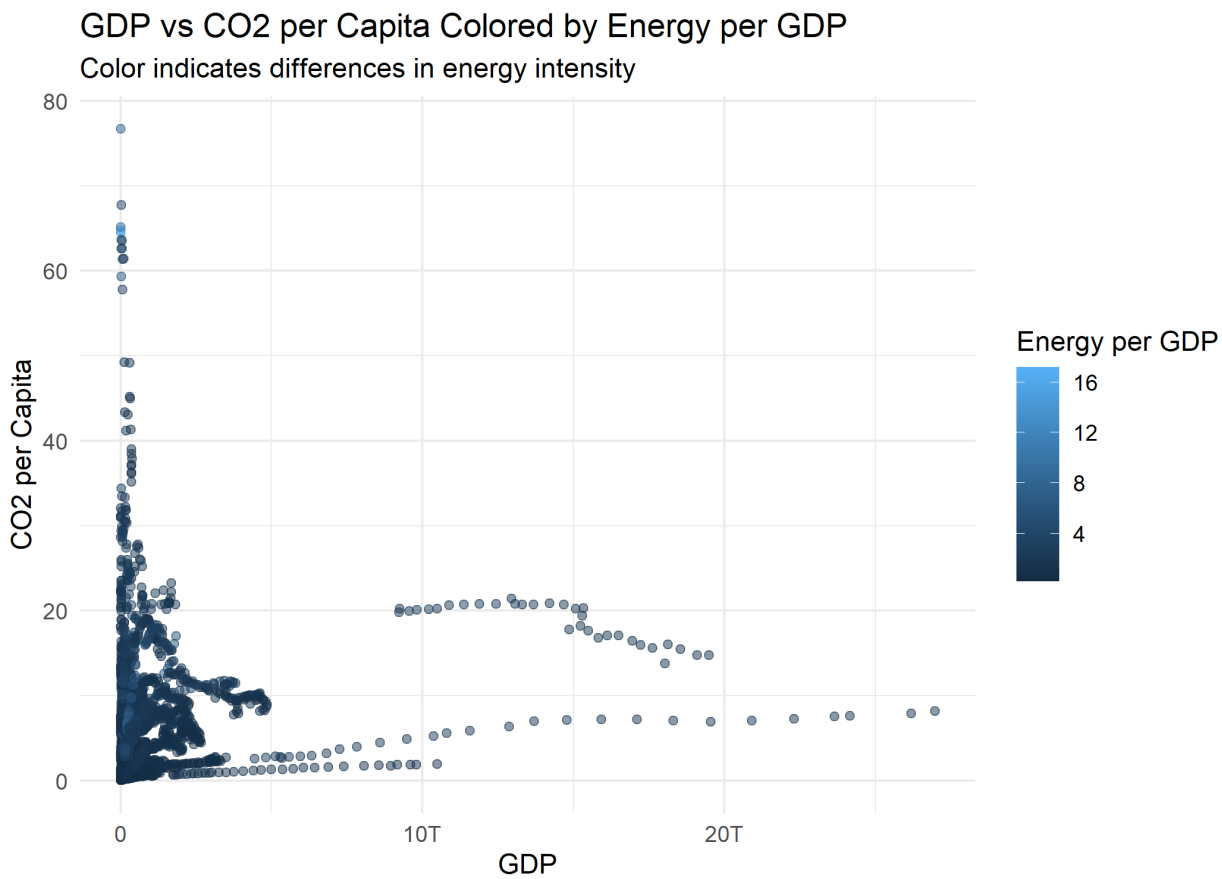
```
ggplot(df_r4, aes(x = energy_per_gdp, y = co2_per_capita)) +
  geom_point(alpha = 0.35, color = "darkgreen") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(
    title = "Energy per GDP vs CO2 per Capita",
    subtitle = "Higher energy intensity is associated with higher emissions",
    x = "Energy per GDP",
    y = "CO2 per Capita"
  ) +
  theme_minimal()
```



3.3 GDP vs CO2 per Capita Colored by Energy Intensity

This plot helps show whether countries with similar GDP levels have different CO2 outcomes depending on energy intensity.

```
ggplot(df_r4, aes(x = gdp, y = co2_per_capita, color = energy_per_gdp)) +
  geom_point(alpha = 0.5) +
  scale_x_continuous(labels = label_number(scale_cut = cut_short_scale())) +
  labs(
    title = "GDP vs CO2 per Capita Colored by Energy per GDP",
    subtitle = "Color indicates differences in energy intensity",
    x = "GDP",
    y = "CO2 per Capita",
    color = "Energy per GDP"
  ) +
  theme_minimal()
```



4 Aggregate Trends Over Time

We summarize yearly averages to study long-run patterns.

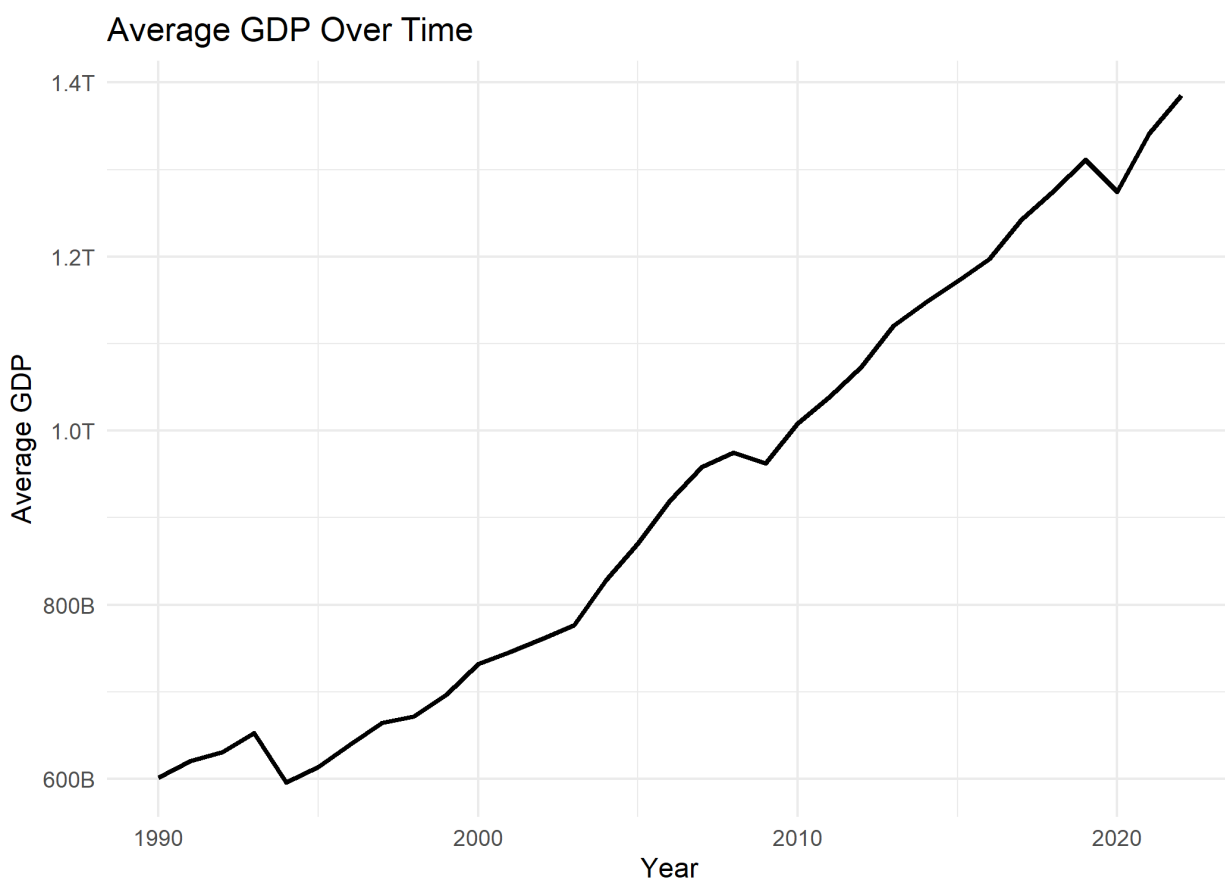
```
df_year <- df_r4 %>%
  group_by(year) %>%
  summarise(
    avg_gdp = mean(gdp, na.rm = TRUE),
    avg_energy_per_gdp = mean(energy_per_gdp, na.rm = TRUE),
    avg_co2_per_capita = mean(co2_per_capita, na.rm = TRUE),
    .groups = "drop"
  )

kable(head(df_year), digits = 3)
```

year	avg_gdp	avg_energy_per_gdp	avg_co2_per_capita
1990	601551378678	2.462	8.902
1991	620612405619	2.446	8.578
1992	630846490933	2.465	8.518

year	avg_gdp	avg_energy_per_gdp	avg_co2_per_capita
1993	652465942774	2.356	8.609
1994	595937287134	2.451	7.885
1995	613564268310	2.376	7.641

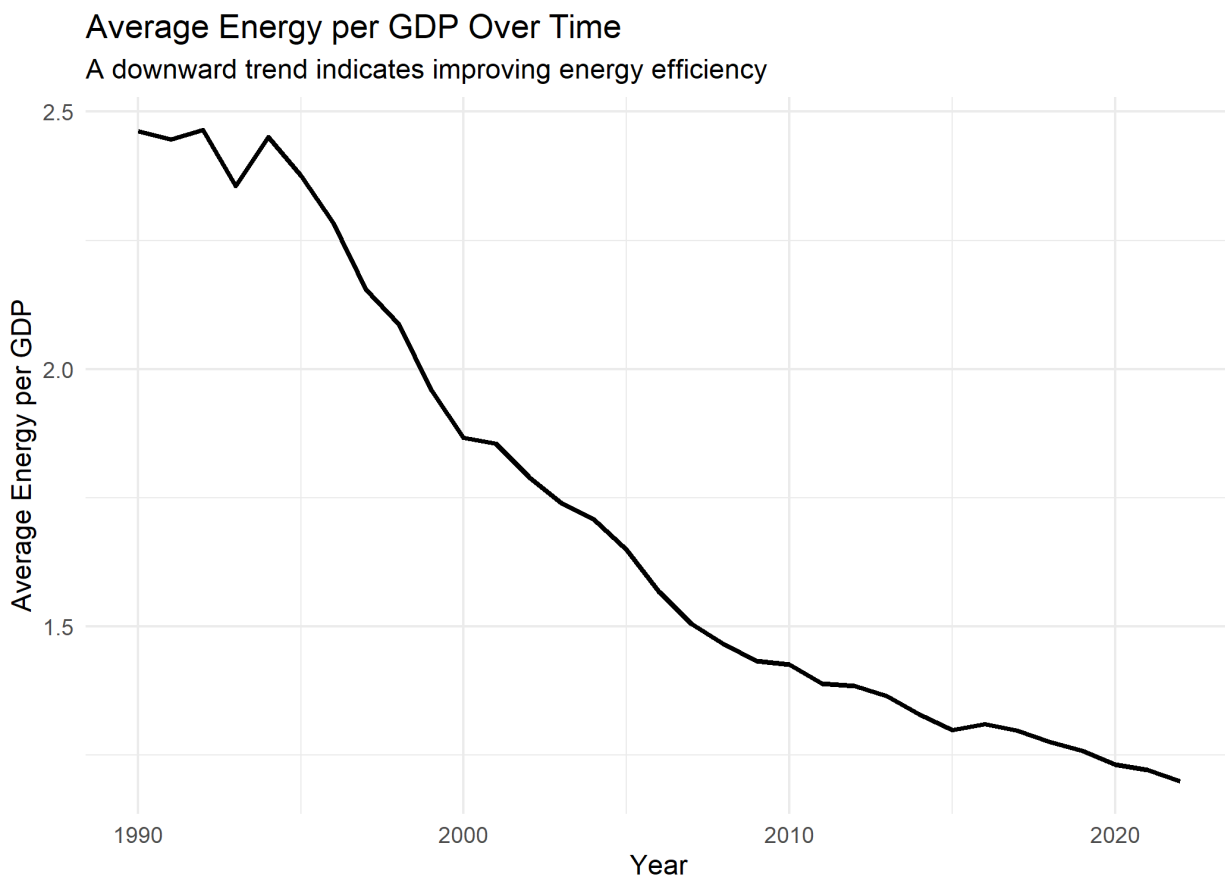
4.1 Average GDP Over Time

```
ggplot(df_year, aes(x = year, y = avg_gdp)) +
  geom_line(linewidth = 1) +
  scale_y_continuous(labels = label_number(scale_cut = cut_short_scale())) +
  labs(
    title = "Average GDP Over Time",
    x = "Year",
    y = "Average GDP"
  ) +
  theme_minimal()
```



4.2 Average Energy per GDP Over Time

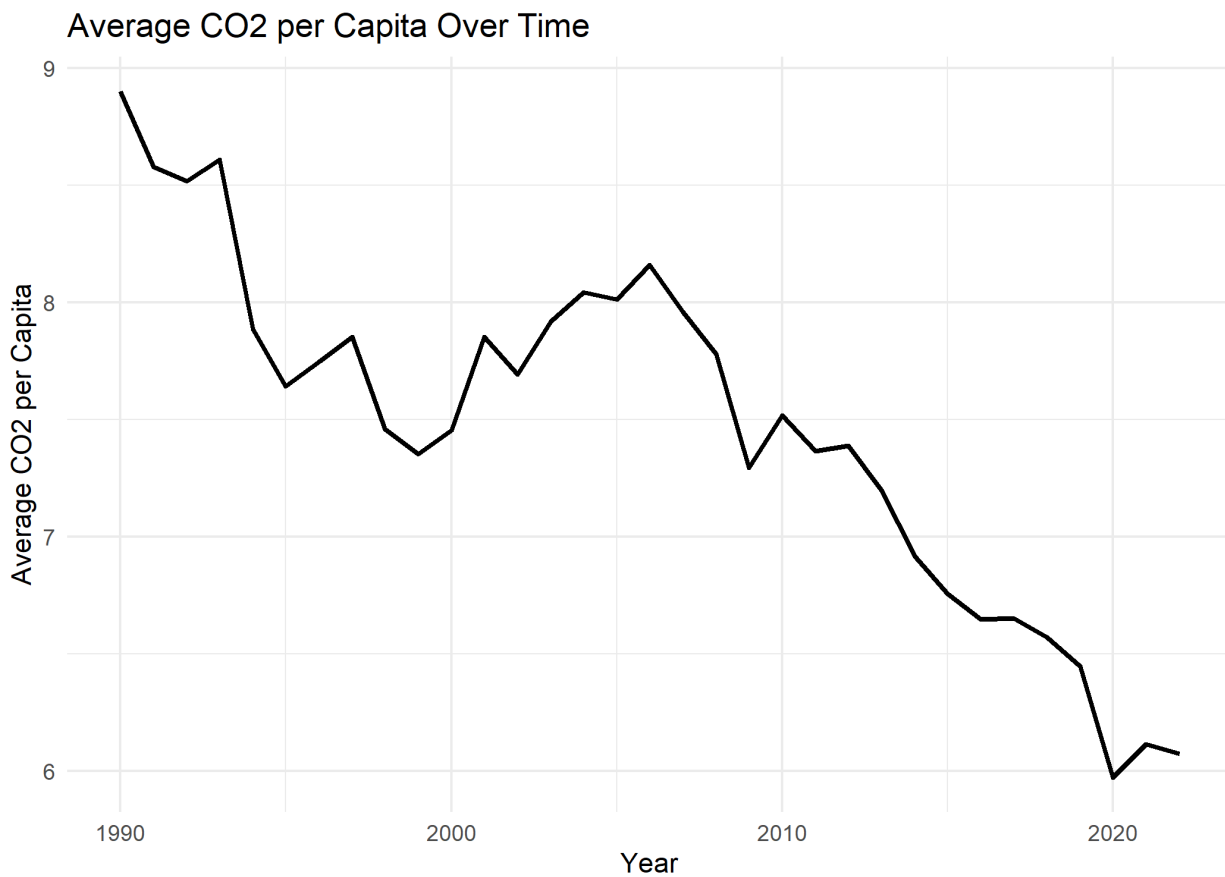
```
ggplot(df_year, aes(x = year, y = avg_energy_per_gdp)) +
  geom_line(linewidth = 1) +
  labs(
    title = "Average Energy per GDP Over Time",
    subtitle = "A downward trend indicates improving energy efficiency",
    x = "Year",
    y = "Average Energy per GDP"
  ) +
  theme_minimal()
```



4.3 Average CO2 per Capita Over Time

```
ggplot(df_year, aes(x = year, y = avg_co2_per_capita)) +
  geom_line(linewidth = 1) +
  labs(
    title = "Average CO2 per Capita Over Time",
    x = "Year",
```

```
    y = "Average CO2 per Capita"
  ) +
  theme_minimal()
```



5 Model 1: GDP and Energy Intensity

This model examines whether GDP and energy intensity are associated with CO2 emissions per capita.

$$CO2_per_capita = \beta_0 + \beta_1 GDP + \beta_2 Energy_per_GDP + \varepsilon$$

```
model1 <- lm(co2_per_capita ~ gdp + energy_per_gdp, data = df_r4)

summary(model1)
```

```
##
## Call:
## lm(formula = co2_per_capita ~ gdp + energy_per_gdp, data = df_r4)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -20.597  -3.411  -1.083   2.056  53.720
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.403e+00  2.511e-01   5.589 2.51e-08 ***
## gdp            3.409e-13  5.430e-14   6.278 3.99e-10 ***
## energy_per_gdp 3.392e+00  1.251e-01  27.126  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.552 on 2637 degrees of freedom
## Multiple R-squared:  0.2254, Adjusted R-squared:  0.2249
## F-statistic: 383.8 on 2 and 2637 DF,  p-value: < 2.2e-16
```

```r
tidy(model1)
```

```
## # A tibble: 3 x 5
##   term           estimate std.error statistic   p.value
##   <chr>             <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)    1.40e+ 0  2.51e- 1      5.59 2.51e-  8
## 2 gdp            3.41e-13  5.43e-14      6.28 3.99e- 10
## 3 energy_per_gdp 3.39e+ 0  1.25e- 1     27.1  3.99e-143
```

6 Model 2: Add Controls

This model adds population and energy use per capita as controls.

$$CO2_per_capita = \beta_0 + \beta_1 GDP + \beta_2 Energy_per_GDP + \beta_3 Population + \beta_4 Energy_per_capita + \varepsilon$$

```r
model2 <- lm(co2_per_capita ~ gdp + energy_per_gdp + population + energy_per_capita, data = df_

summary(model2)
```

```
##
## Call:
## lm(formula = co2_per_capita ~ gdp + energy_per_gdp + population +
##     energy_per_capita, data = df_r4)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -22.604  -0.677   0.041   0.768  32.667
##
```

```
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -4.768e-01  1.331e-01  -3.582 0.000347 ***
## gdp                1.749e-13  3.944e-14   4.434 9.63e-06 ***
## energy_per_gdp     7.834e-01  7.193e-02  10.891  < 2e-16 ***
## population        -7.816e-10  4.853e-10  -1.611 0.107396
## energy_per_capita  1.706e-04  2.102e-06  81.161  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.389 on 2635 degrees of freedom
## Multiple R-squared:  0.7929, Adjusted R-squared:  0.7926
## F-statistic:  2522 on 4 and 2635 DF,  p-value: < 2.2e-16
```

```r
tidy(model2)
```

```
## # A tibble: 5 x 5
##   term               estimate std.error statistic  p.value
##   <chr>                 <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)       -4.77e- 1  1.33e- 1     -3.58 3.47e- 4
## 2 gdp                1.75e-13  3.94e-14      4.43 9.63e- 6
## 3 energy_per_gdp     7.83e- 1  7.19e- 2     10.9  4.77e-27
## 4 population        -7.82e-10  4.85e-10     -1.61 1.07e- 1
## 5 energy_per_capita  1.71e- 4  2.10e- 6     81.2  0
```

7 Model 3: Interaction Between GDP and Energy Intensity

This model tests whether the effect of GDP on CO2 emissions depends on energy intensity.

$$CO2_per_capita = \beta_0 + \beta_1 GDP + \beta_2 Energy_per_GDP + \beta_3 (GDP \times Energy_per_GDP) + \varepsilon$$

```r
model3 <- lm(co2_per_capita ~ gdp * energy_per_gdp, data = df_r4)

summary(model3)
```

```
##
## Call:
## lm(formula = co2_per_capita ~ gdp * energy_per_gdp, data = df_r4)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -20.732  -3.397  -1.089   2.037  53.673
##
## Coefficients:
```

```
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         1.349e+00  2.727e-01   4.947 8.01e-07 ***
## gdp                 4.534e-13  2.273e-13   1.995   0.0461 *
## energy_per_gdp      3.421e+00  1.370e-01  24.963  < 2e-16 ***
## gdp:energy_per_gdp -6.558e-14  1.287e-13  -0.510   0.6103
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.553 on 2636 degrees of freedom
## Multiple R-squared:  0.2255, Adjusted R-squared:  0.2246
## F-statistic: 255.8 on 3 and 2636 DF,  p-value: < 2.2e-16
```

```r
tidy(model3)
```

```
## # A tibble: 4 x 5
##   term                estimate std.error statistic   p.value
##   <chr>                  <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)         1.35e+ 0  2.73e- 1     4.95  8.01e-  7
## 2 gdp                 4.53e-13  2.27e-13     1.99  4.61e-  2
## 3 energy_per_gdp      3.42e+ 0  1.37e- 1    25.0   1.21e-123
## 4 gdp:energy_per_gdp -6.56e-14  1.29e-13    -0.510 6.10e-  1
```

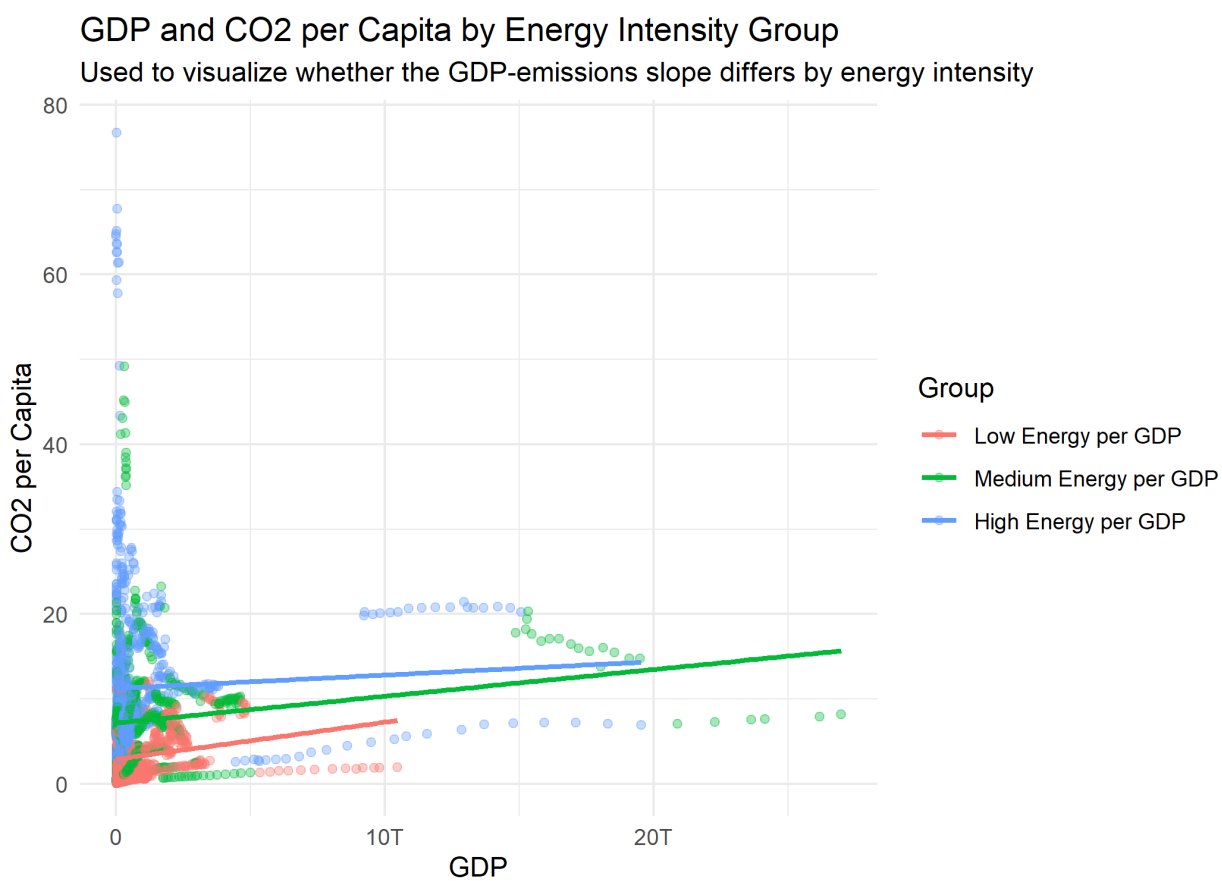
7.1 Interaction Visualization

```r
df_r4_grouped <- df_r4 %>%
  mutate(
    intensity_group = ntile(energy_per_gdp, 3),
    intensity_group = factor(
      intensity_group,
      levels = c(1, 2, 3),
      labels = c("Low Energy per GDP", "Medium Energy per GDP", "High Energy per GDP")
    )
  )

table(df_r4_grouped$intensity_group)
```

```
##
##    Low Energy per GDP Medium Energy per GDP   High Energy per GDP
##                   880                   880                   880
```

```r
ggplot(df_r4_grouped, aes(x = gdp, y = co2_per_capita, color = intensity_group)) +
  geom_point(alpha = 0.35) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_x_continuous(labels = label_number(scale_cut = cut_short_scale())) +
```

```
  labs(
    title = "GDP and CO2 per Capita by Energy Intensity Group",
    subtitle = "Used to visualize whether the GDP-emissions slope differs by energy intensity"
    x = "GDP",
    y = "CO2 per Capita",
    color = "Group"
  ) +
  theme_minimal()
```



8 Model 4: Fixed Effects Panel Model

A fixed effects model compares each country to itself over time. This is stronger than pooled OLS because it controls for time-invariant country characteristics such as geography, industrial structure, and long-run institutional differences.

```
pdata <- pdata.frame(df_r4, index = c("country", "year"))

panel_model <- plm(
  co2_per_capita ~ log(gdp) + energy_per_gdp,
  data = pdata,
```

```r
  model = "within",
  effect = "individual"
)

summary(panel_model)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = co2_per_capita ~ log(gdp) + energy_per_gdp, data = pdata,
##     effect = "individual", model = "within")
##
## Unbalanced Panel: n = 91, T = 2-33, N = 2640
##
## Residuals:
##       Min.    1st Qu.     Median    3rd Qu.       Max.
## -14.000338  -0.439461   0.046842   0.598352  22.141147
##
## Coefficients:
##                Estimate Std. Error t-value  Pr(>|t|)
## log(gdp)       0.765590   0.114029   6.714 2.327e-11 ***
## energy_per_gdp 1.378171   0.064554  21.349 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    10157
## Residual Sum of Squares: 8412.7
## R-Squared:      0.17171
## Adj. R-Squared: 0.14179
## F-statistic: 264.007 on 2 and 2547 DF, p-value: < 2.22e-16
```

9 Model 5: Environmental Kuznets Curve

This model tests whether the GDP-emissions relationship is nonlinear by including a squared GDP term.

$$CO2_per_capita = \beta_0 + \beta_1 GDP + \beta_2 GDP^2 + \beta_3 Energy_per_GDP + \varepsilon$$

```r
model_ekc <- lm(co2_per_capita ~ gdp + I(gdp^2) + energy_per_gdp, data = df_r4)

summary(model_ekc)
```

```
##
## Call:
## lm(formula = co2_per_capita ~ gdp + I(gdp^2) + energy_per_gdp,
```

```
##     data = df_r4)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -20.547  -3.345  -1.112   2.033  53.826
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     1.219e+00  2.634e-01   4.629 3.86e-06 ***
## gdp             6.229e-13  1.342e-13   4.642 3.61e-06 ***
## I(gdp^2)       -1.773e-26  7.717e-27  -2.298   0.0217 *
## energy_per_gdp  3.411e+00  1.252e-01  27.239  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.547 on 2636 degrees of freedom
## Multiple R-squared:  0.227,  Adjusted R-squared:  0.2261
## F-statistic:   258 on 3 and 2636 DF,  p-value: < 2.2e-16
```

```r
tidy(model_ekc)
```

```
## # A tibble: 4 x 5
##   term            estimate std.error statistic   p.value
##   <chr>              <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)     1.22e+ 0  2.63e- 1      4.63 3.86e-  6
## 2 gdp             6.23e-13  1.34e-13      4.64 3.61e-  6
## 3 I(gdp^2)       -1.77e-26  7.72e-27     -2.30 2.17e-  2
## 4 energy_per_gdp  3.41e+ 0  1.25e- 1     27.2  3.63e-144
```

10 Diagnostics

10.1 Multicollinearity

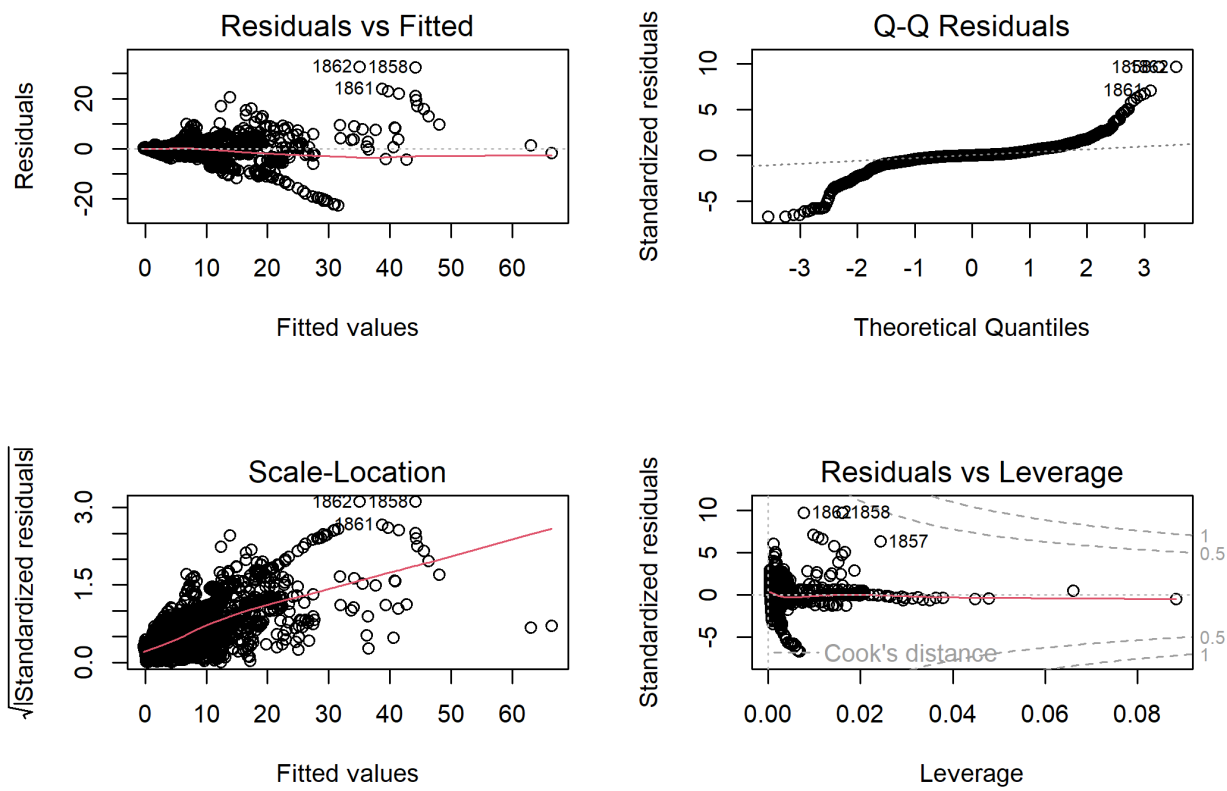
```r
vif(model2)
```

```
##               gdp    energy_per_gdp        population energy_per_capita
##          1.973113          1.237129          2.002500          1.336083
```

VIF values below 5 suggest that multicollinearity is not a major concern in this model.

10.2 Residual Diagnostics

```r
par(mfrow = c(2, 2))
plot(model2)
```




```r
par(mfrow = c(1, 1))
```

The diagnostic plots should be interpreted as follows:

- Residuals vs Fitted: checks linearity and mean-zero residuals
- Normal Q-Q: checks whether residuals are approximately normal
- Scale-Location: checks whether variance is approximately constant
- Residuals vs Leverage: identifies influential observations

11 Model Comparison

We compare the main OLS-style models using adjusted R-squared and AIC.

```r
comparison <- tibble(
  Model = c(
    "Model 1: GDP + Energy per GDP",
```

```r
    "Model 2: Add Controls",
    "Model 3: Interaction",
    "Model 5: EKC"
  ),
  Adjusted_R2 = c(
    summary(model1)$adj.r.squared,
    summary(model2)$adj.r.squared,
    summary(model3)$adj.r.squared,
    summary(model_ekc)$adj.r.squared
  ),
  AIC = c(
    AIC(model1),
    AIC(model2),
    AIC(model3),
    AIC(model_ekc)
  )
)

kable(comparison, digits = 3)
```

Model	Adjusted_R2	AIC
Model 1: GDP + Energy per GDP	0.225	17422.17
Model 2: Add Controls	0.793	13943.94
Model 3: Interaction	0.225	17423.92
Model 5: EKC	0.226	17418.89

12 Results and Interpretation

The results show that GDP is positively associated with CO2 emissions per capita, but its explanatory power is limited when used only with energy intensity. Model 1 explains only a modest share of the overall variation in emissions.

After adding controls in Model 2, the fit improves substantially. This indicates that direct energy use per capita is a major determinant of CO2 emissions. In the full model, energy-related variables are much more informative than population, and population is not statistically significant once the other variables are included.

Energy intensity remains statistically significant across the main specifications. This means that economies using more energy per unit of output tend to have higher CO2 emissions per capita, even after accounting for GDP.

The interaction model does not provide evidence that the effect of GDP changes meaningfully with energy intensity. The interaction term is not statistically significant, so the data do not support the claim that energy intensity strongly moderates the GDP-emissions relationship in this specification.

The EKC model provides evidence of a nonlinear relationship between GDP and CO2 emissions. The positive coefficient on GDP and the negative coefficient on the squared GDP term are consistent

with an inverted-U pattern, which suggests that emissions may rise in earlier stages of development and level off or decline at higher development levels.

The fixed effects model confirms that the positive association of log(GDP) and energy intensity with CO2 emissions also holds within countries over time

13 Conclusion

This analysis finds that economic development, energy intensity, and energy use are all related to CO2 emissions, but their roles are not equally important. GDP is positively associated with emissions, yet the strongest predictor in the expanded model is energy use per capita. This suggests that emissions are driven more by how much energy countries consume than by economic size alone.

Energy intensity is also an important predictor. Countries that require more energy to generate output tend to have higher CO2 emissions per capita. This supports the idea that improving efficiency and reducing energy intensity can contribute to lower emissions.

At the same time, the interaction model shows that the effect of GDP on emissions does not significantly change across different levels of energy intensity in this specification. Therefore, the data do not support a strong moderation effect here.

The Environmental Kuznets Curve model provides evidence of a nonlinear GDP-emissions relationship. The pattern is consistent with emissions rising during earlier stages of development and weakening at higher levels of development.

Overall, the main implication of this section is that policies targeting energy use and energy efficiency are likely to be more effective for reducing emissions than policies focused only on slowing economic growth.

14 Limitations

This analysis uses observational data, so the results should be interpreted as associations rather than causal effects.

The pooled OLS models may still be influenced by omitted variables, although the fixed effects model reduces this problem by controlling for unobserved country-level characteristics that are constant over time.

The relationship between GDP and emissions may be more complex than the linear and quadratic forms estimated here, and future work could evaluate more flexible nonlinear methods.

Finally, the dataset is unbalanced across countries and years, and measurement differences across countries may also affect the estimates.